

TrES-3b

R-0284

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180512 · created 2026-07-17T16:06:26+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458250.7605 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.033
Transit depth (Rp/Rs)^2	2.97 +/- 1.14 [%]
Orbital Inclination (inc)	81.71 +/- 0.92
Airmass coefficient 1 (a1)	0.987 +/- 0.015
Airmass coefficient 2 (a2)	0.0083 +/- 0.0057
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	#1 - [301, 255]
Optimal Aperture	3.96
Optimal Annulus	8.6
Transit Duration (day)	0.053 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.172 $\pm$ 0.033 vs 0.16309 (deviation 0.26 σ)
Duration fit vs published	0.053 d vs 0.0567 d (deviation 0.33 σ)
Mid-time O-C	-4.2 min
Depth SNR	5.1
Residual scatter	1.48 %

Light curve

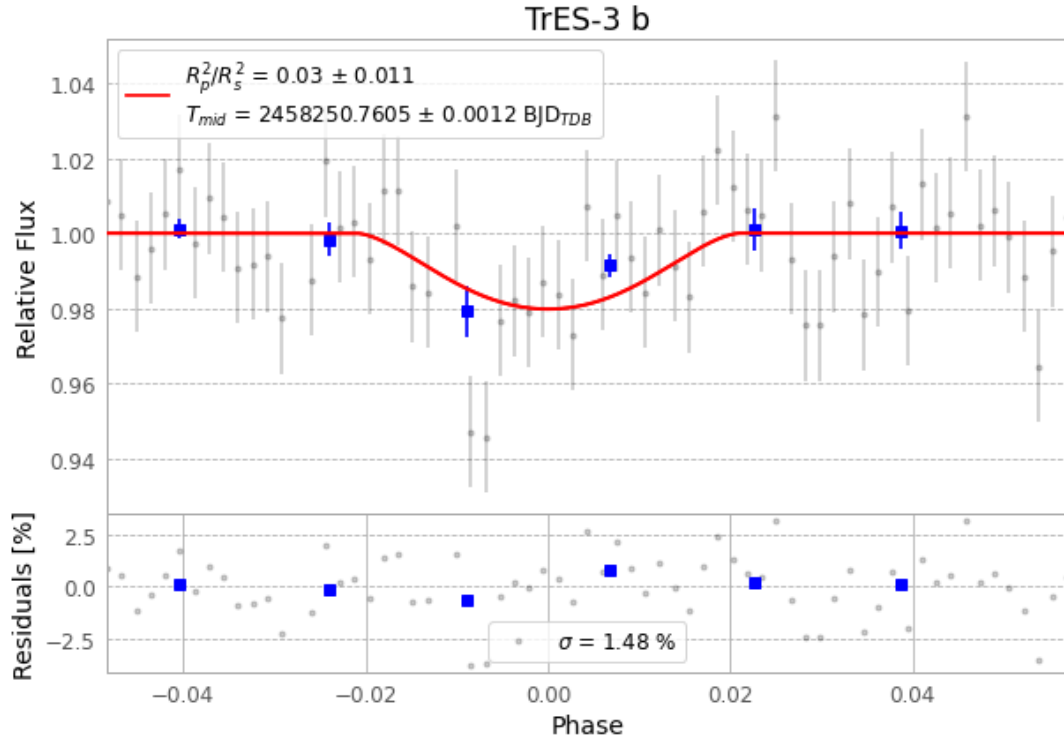


Figure 1 — FinalLightCurve_TrES-3 b_2018-05-11.png (SHA-256 2fdebfe93ae05e32...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 170], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 98ffc348ed92f737...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

67 FITS frames (44 MB), TRES-3180512043912.FITS → TRES-3180512075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180512.log (463faae100c9...), FinalLightCurve_TrES-3 b_2018-05-11.png (2fdebfe93ae0...), FinalLightCurve_TrES-3 b_2018-05-11.csv (2269c3beaa9d...)

Compute resources

Wall time 230.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 16:06Z.